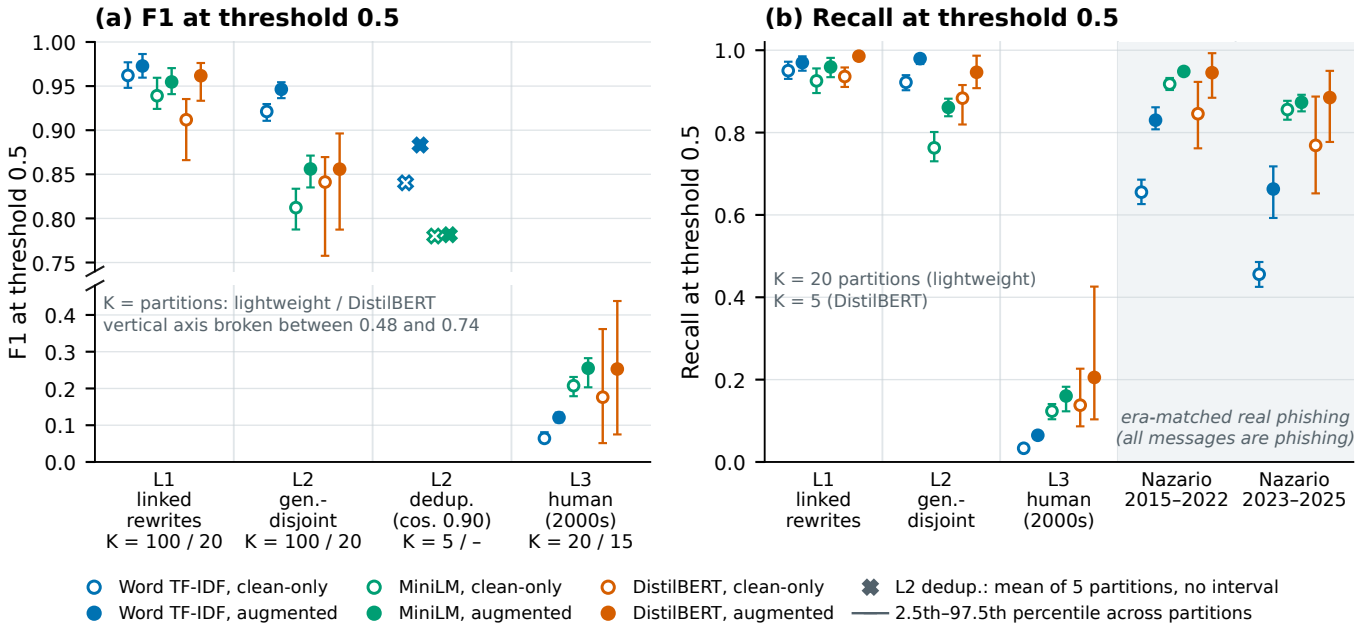
(a) F1 at threshold 0.5
F1 at threshold 0.5
1.00
0.95
0.90
0.85
0.80
0.75
0.4
0.3
0.2
0.1
0.0
K = partitions: lightweight / DistilBERT
vertical axis broken between 0.48 and 0.74
L1 linked rewrites K = 100 / 20
L2 gen.-disjoint K = 100 / 20
L2 dedup. (cos. 0.90) K = 5 / –
L3 human (2000s) K = 20 / 15
(b) Recall at threshold 0.5
Recall at threshold 0.5
1.0
0.8
0.6
0.4
0.2
0.0
K = 20 partitions (lightweight)
K = 5 (DistilBERT)
era-matched real phishing
(all messages are phishing)
L1 linked rewrites
L2 gen.-disjoint
L3 human (2000s)
Nazario 2015–2022
Nazario 2023–2025
Word TF-IDF, clean-only
Word TF-IDF, augmented
MiniLM, clean-only
MiniLM, augmented
DistilBERT, clean-only
DistilBERT, augmented
L2 dedup.: mean of 5 partitions, no interval
2.5th–97.5th percentile across partitions